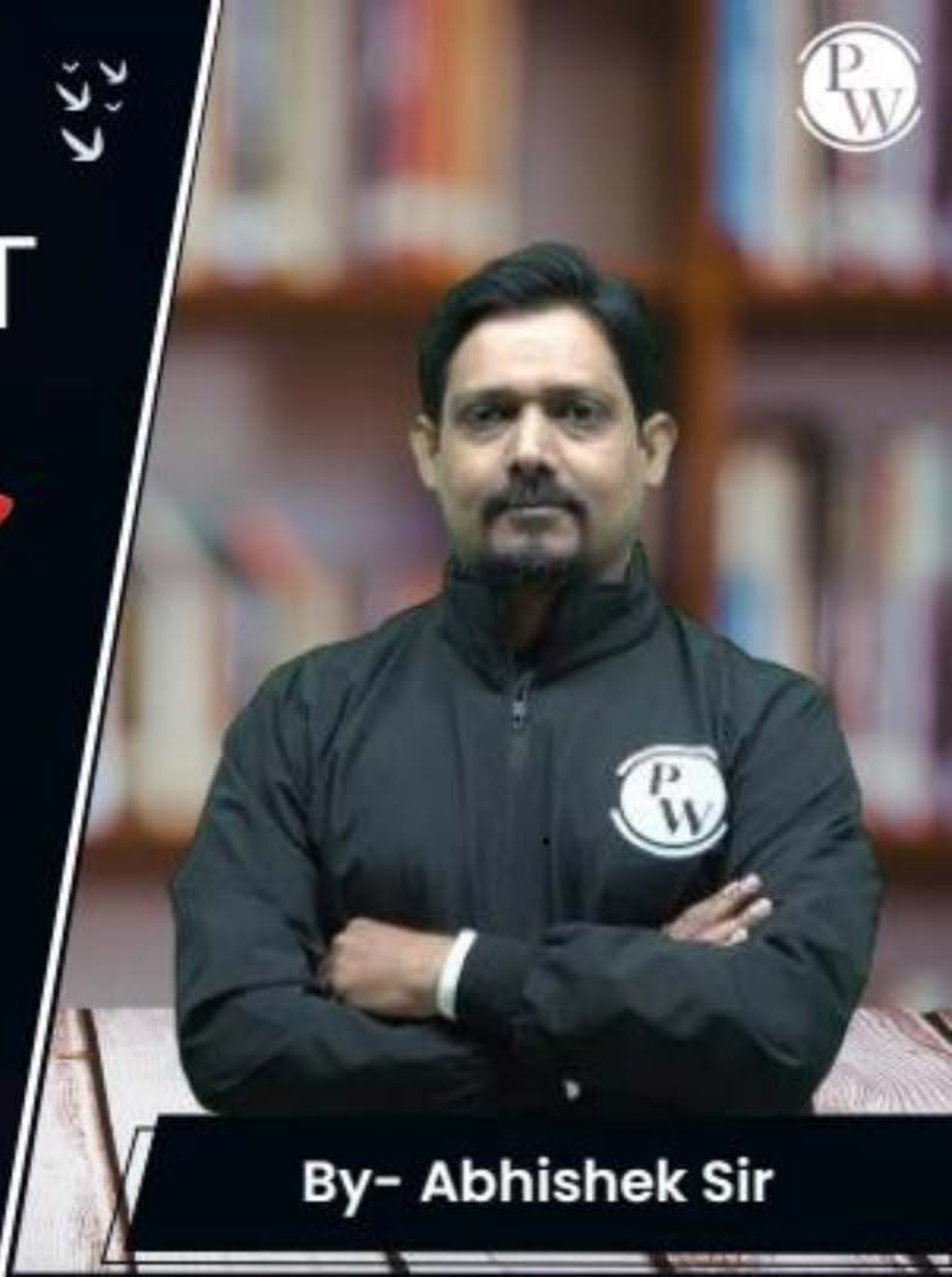
Computer Science & IT



C programming

Data Types & Operators

Lecture No. 03



By- Abhishek Sir

Recap of Previous Lecture



- topic
- topic
- topic
- topic
- topic

int

unsigned ("%u"); 1111 ⇒ (15)

Signed ("%d"); 1111 = (-1)

Short (2B), long (8B)

4bit ↓

Slide

topics to be Covered



- topic
- topic
- topic
- topic
- topic

character & float

Arithmetic operator

Slide



Topic : Agenda

Character
Operator
Precedence & Associativity
Expression
++ and - -



Character

```
char ch;
```

```
char ch = 'a';
```
declaration & initialization

```
printf("%c", ch);
```
output a without an quotes



Character

ASCII Encoding

ch is of 1 Byte

for every character one Numeric value assigned

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ASCII value

0...9	48...57
A...Z	65...90
a...z	97...122

Slide



Character

```c
#include <stdio.h>
#include <stdio.h>
int main(){
        char ch1 = 'a';
        char ch2 = 'z';
        printf("%c\n", ch1);
        printf("%d\n", ch1);
        printf("%c\n", ch2);
        printf("%d", ch2);
return 0;
}
```

a

ASCII value. 97

z

ASCII value 122

Slide

```c
#include<stdio.h>
int main() {
    char ch = 'b';
    printf("%c", ch+5); // 98+5=103
    return 0;
}
```

Character

ASCII value

character

g output

a - 97
b - 98
c - 99
d - 100
e - 101
f - 102
g - 103



Character

```c
#include <stdio.h>
#include <stdio.h>
int main(){
        char ch1 = 'A';
        char ch2 = 'Z';
        printf("%c\n", ch1);
        printf("%d\n", ch1);
        printf("%c\n", ch2);
        printf("%d", ch2);
return 0;
}
```



Character

```c
#include <stdio.h>
#include <stdio.h>
int main(){
        char ch1 = '0';
        char ch2 = '9';
        printf("%c\n", ch1);
        printf("%d\n", ch1);
        printf("%c\n", ch2);
        printf("%d", ch2);
        return 0;
}
```



Character

```c
#include <stdio.h>
int main(){
        char a = 128;
        printf("%c\n", a);
        printf("%d\n", a);
        return 0;
}
```



ASCII Table

0	NUL	16	DLE	32		48	0	64	@	80	P	96	`	112	p
1	SOH	17	DC1	33	!	49	1	65	A	81	Q	97	a	113	q
2	STX	18	DC2	34	"	50	2	66	B	82	R	98	b	114	r
3	ETX	19	DC3	35	#	51	3	67	C	83	S	99	c	115	s
4	EOT	20	DC4	36	$	52	4	68	D	84	T	100	d	116	t
5	ENQ	21	NAK	37	%	53	5	69	E	85	U	101	e	117	u
6	ACK	22	SYN	38	&	54	6	70	F	86	V	102	f	118	v
7	BEL	23	ETB	39	'	55	7	71	G	87	W	103	g	119	w
8	BS	24	CAN	40	(	56	8	72	H	88	X	104	h	120	x
9	HT	25	EM	41	)	57	9	73	I	89	Y	105	i	121	y
10	LF	26	SUB	42	*	58	:	74	J	90	Z	106	j	122	z
11	VT	27	ESC	43	+	59	;	75	K	91	[	107	k	123	{
12	FF	28	FS	44	,	60	<	76	L	92	\	108	l	124	|
13	CR	29	GS	45	-	61	=	77	M	93	]	109	m	125	}
14	SO	30	RS	46	.	62	>	78	N	94	^	110	n	126	~
15	SI	31	US	47	/	63	?	79	O	95	_	111	o	127	DEL

6+5

98+5

=103

Slide



Topic : Operators & Arithmetic Expresion

+, −, *, /, =, %

① 5+6*7 = 5+42 = 47

② 5*6/2

5*3

= 15

BODMAS

Any arithmetic expression required order of evaluation

Slide



Topic : Arithmetic Expression Rule

Precedence :- order of evaluation of operator decided by precedence Rule.

Associativity :- if precedence of operator is same then we use associativity rule for order of evaluation.

Slide



Topic : Operators



$-5+6*2$ → unary minus

$-5+12$

-17 Wrong

unary precedence higher than binary

Single variable or

Single numeric

$-5+12$

$=7$

Slide



Topic : Operators

$(5+6)+7$

Govt job ↑ private

$= 11+7$

$= 18$

Same precedence

Left Associative

C Language +, − have same precedence

$5-6+7 = -1+7 = 6$

↑

Slide



Topic : Operators

= operator

Right Associative

```
int a, b=10, c=20;
a=b=c;
```

right Associative

private ↑ Govt

a=(b=c)

a = 20 (2)
b = 20
c = 20 (1)

Slide



Topic : Operators

$(-4)+7$

$-4+7 = 3$

-4 ←

↑ unary

$4+(-6)$

Slide



Topic : Arithmetic Operator

+, -, *, /, %

Slide



Topic : Arithmetic Operator

Dividend
Divisor
Quotient
Remainder

Dividend = Quotient * Divisor+ Remainder



Topic : Arithmetic Operator

7 / 3,
-7/-3

-7/3,

7/-3,

7 ← dividend

3 ← divisor

2 - Quotient if two integer divided then quotient Ans

1 - Remainder 7%3 then Answer is remainder

Slide

$\frac{7}{3}$ ←

Q : 2

R : 1

$\frac{7}{-3}$ ←

Q -2

R 1

$\frac{-7}{3}$

Q -2

R -1

$\frac{-7}{-3}$

Q 2

R -1

$-3 \times 2 - 1$

$-6 - 1 = -7$

Dividend = quotient * divisor + Remainder

Dividend +ve

Remainder +ve

Dividend -ve

Remainder Negative



Program

```
#include <stdio.h>
int main() {
printf("%d\n",7/3);
printf("%d\n",7/-3);
printf("%d\n",-7/3);
printf("%d\n",-7/-3);
printf("%d\n",7%3);
printf("%d\n",7%-3);
printf("%d\n",-7%3);
printf("%d\n",-7%-3);
   return 0;
}
```



Precedence & Associativity Rule

bracket	( )		Highest
Unary minus	-		
Multiplicative	*, / , %	Left to Right	
Additive	+ -	Left to Right	
Equal	=	Right to Left	Lowest

Left Associativity

Left Associative

Right Associative

Slide



Precedence & Associativity Rule

* / %	Multiplication/division/modulus	left-to-right
+ -	Addition/subtraction	left-to-right
<< >>	Bitwise shift left, Bitwise shift right	left-to-right
< <= > >=	Relational less than/less than or equal to Relational greater than/greater than or equal to	left-to-right
== !=	Relational is equal to/is not equal to	left-to-right
&	Bitwise AND	left-to-right
^	Bitwise exclusive OR	left-to-right
|	Bitwise inclusive OR	left-to-right
&&	Logical AND	left-to-right
||	Logical OR	left-to-right



Precedence & Associativity Rule

? :	Ternary conditional	right-to-left
=	Assignment	right-to-left
+= -=	Addition/subtraction assignment	
*= /=	Multiplication/division assignment	
%= &=	Modulus/bitwise AND assignment	
^= |=	Bitwise exclusive/inclusive OR assignment	
<<= >>=	Bitwise shift left/right assignment	
,	Comma (separate expressions)	left-to-right

Arithmetic operation between int & int int $\frac{5}{2} = 2$

Arithmetic operation between int float ans is float

$$\frac{5}{2.0} = 2.5$$

Arithmetic operation between float & float is float

$$\frac{5.0}{2.0} = 2.5$$



Toipc:Question

```
#include <stdio.h>
int main(void){
float x;
x = 7*2.0/2+10/3;
printf("%f", x);
return 0;
}
```

The value of x is ___

A. 10

B. 10.0

C. 10.33

D. 11.0

$X = 7*2.0/2 + 10/3$

$= 14.0/2 + 3$

$= 7.0 + 3 = 10.0$

Slide



Toipc:Question

```
#include<stdio.h>
   int main(){
   int x;
   x= -2 + 11 - 7 * 9 % 6 / 12;
   printf("%d",x);
   return 0 ;
   }
```

The value of x is

____

A. 6
B. 7
C. 8
D. 9

-2+11-7*9%6/12

-2+11-63%6/12

=-2+11-3/12

-2+11-0

-2+11 = 9

Slide



Toipc:Question

3/12 Q: 0 R: 3

```
#include<stdio.h>
   int main(){
   int x;
   x=   2 * 3/4+4/4 +8-2+5/8;
   printf("%d",x);
   return 0 ;
   }
```

The value of x is

(A) 6
(b) 7
(c) 8
(d) 9

2*3/4+4/4+8-2+5/8

6/4+1+8-2+0

1+1+8-2 = 2+8-2 = 10-2 = 8

Slide



Toipc:Question

```
#include<stdio.h>
   int main(){
   int x;
   x=  3/2*4+3/8+3;
   printf("%d",x);
   return 0 ;
   }
```

The value of x is

____

(a) 6
(b) 7
(c) 8
(d) 9

HW

Slide



Toipc:Question

```
#include<stdio.h>
   int main(){
   int x;
   x=  -14%8+2%-8;
   printf("%d",x);
   return 0 ;
   }
```

The value of x is

____

(a) -6
(b) 7
(c) -4
(d) 9

HW

Slide



2 mins Summary

topic — Character

topic — Arithmetic operator

topic — precedence & Associativity

topic — Division / Modulo

topic

Slide

THANK - YOU